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Investor Memory and Biased Beliefs: Evidence from the Field

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1 Big picture

- **Question.** How does investor memory shape return expectations and trading decisions? More specifically, do retail investors selectively recall past market and portfolio experiences, and do these recalled experiences explain biased beliefs?
- **Setting.** The paper studies a large, nationally representative sample of Chinese retail investors. The authors elicit investors' memories of stock-market experiences and their return expectations, and for a subsample they merge the survey responses to administrative transaction records.
- **Main idea.** Investors do not form beliefs only by mechanically averaging objective past returns. They retrieve memories. Retrieval is selective, cued by the current environment, and affected by similarity, recency, and salience. These retrieved memories are then used to simulate future outcomes.
- **Core mechanism.** A rising current market can cue investors to recall more positive past experiences. Positive recalls then lead to more optimistic expected returns and lower perceived crash probabilities. In this way, memory provides a microfoundation for extrapolative beliefs.
- **Conceptual framework.** The paper adapts the memory-based belief-formation model of Bordalo et al. (2025). The model has two steps:
 - **recall:** the investor retrieves past experiences, with retrieval probabilities tilted toward experiences similar to the current cue;
 - **simulation:** the investor uses retrieved experiences to form a subjective distribution of future returns.
- **Empirical design.** The survey is rolled out over six weeks. Because investors begin the survey at different dates and times, they face different contemporaneous market-return cues. This timing variation is used to test whether current returns affect recalled experiences.
- **Main data advantage.** The authors observe three objects for many investors:
 - what they *actually* experienced, from market and transaction data;
 - what they *recall* experiencing, from survey questions;
 - what they *expect* and later *do*, from survey beliefs and trading records.
- **Key result 1: memory is real but distorted.** Recalled experiences are correlated with actual experiences, so investors are not answering randomly. But memory is also selective: investors disproportionately recall recent episodes and salient bubble/crash episodes such as 2007–2008 and 2014–2015.
- **Key result 2: memory is cued.** In the ProbedRecall task, a 1 percentage point increase in the survey-day market return is associated with a 52 basis point increase in recalled own return for the previous day and a 113 basis point increase in recalled own return over the past month.
- **Key result 3: recalled experience dominates actual experience.** Recalled returns are strongly related to return expectations. In horse races, recalled experiences generally explain beliefs better than actual market or own returns.
- **Key result 4: recalled beliefs matter for action.** More optimistic investors buy more stocks after the survey. The part of return expectations explained by recall is especially strongly related to net buying.

- **Main contribution.** The paper provides field evidence that memory is not just a laboratory phenomenon. It shapes beliefs and trading behavior in a high-stakes financial market.
- **Important caveat.** The memory-belief link is not fully causal. The paper is careful: recall and belief are elicited in a survey, so anchoring, priming, motivated reasoning, and consistency motives must be considered. The authors run several checks, but they still interpret the memory-to-belief estimates as likely upper bounds for recalled personal returns in this survey environment.

2 Data and measurement (key objects)

2.1 Survey sample and implementation

- **Institutional partner.** The survey is implemented with a large financial institution in China that has relationships with major brokerage firms and access to retail-investor trading records.
- **Sampling design.** The collaborating institution randomizes the survey across branch offices of China’s 60 largest brokers. The number of branch offices selected in each province is proportional to the province’s trading volume.
- **Geographic coverage.** The survey covers 2,993 branch offices across 30 provinces and regions.
- **Survey period.** The survey is conducted between November 29, 2021 and January 9, 2022, in three waves of two weeks each.
- **Initial responses.** The initial survey sample contains 37,921 responses.
- **Quality filters.** A response is dropped if the investor:
 - spent less than 175 seconds on the survey;
 - failed the comprehension checks on dollar investment and dollar return;
 - recalled a FreeRecall episode longer than 10 years.
- **Main sample.** After filters, the main sample has 17,324 valid responses.
- **Merged sample.** For 5,154 investors, survey responses are merged with administrative transaction-level data. The merge requires name and date of birth to identify the same individual uniquely.
- **Representative but not frictionless.** The sample is large and geographically broad, but investors who take the survey may be more connected to brokerage account managers and may have higher net worth than the average retail investor.

2.2 Survey-block design

The survey is organized to elicit recall first and beliefs later. This order is chosen to avoid contaminating the free-recall task with later expectation questions, though it creates possible priming concerns that the paper later tests.

1. **Comprehension block.** Investors answer basic questions about financial concepts such as dollar investment and dollar return. Those who fail are excluded.

2. **FreeRecall block.** Investors are asked to recall a past market episode that comes to mind. There are three randomized treatments:
 - **Neutral:** recall the market episode that first comes to mind;
 - **Happy:** recall the happiest market episode;
 - **Painful:** recall the most painful market episode.

After reporting the episode’s start and end months, investors recall the Shanghai Composite return during that episode and their own RMB investment and RMB return during that episode.
3. **ProbedRecall block.** Investors are asked to recall their own portfolio returns over fixed horizons: yesterday, the past month, the past year, and the past five years.
4. **Expectation block.** Investors report expectations about:
 - market returns over the next month and next year;
 - own portfolio returns over the next month and next year;
 - crash probabilities, such as the Shanghai Composite falling below 3,000 within one month or below 2,500 within one year.
5. **Personality and demographics.** The survey also collects Big Five personality measures, gender, age, education, income, wealth, account-checking frequency, news-checking frequency, investment-discussion frequency, and number of WeChat groups.

2.3 Notation

The paper uses a compact notation. Throughout the note:

- $R[\cdot]$ denotes a recalled value.
- $E[\cdot]$ denotes an expected value or subjective belief.
- MktRet denotes market return, usually the Shanghai Composite return.
- OwnRet denotes the investor’s own portfolio return.
- $R[\text{MktRet}_{\text{episode}}]$ is the recalled market return during the episode elicited in FreeRecall.
- $\text{MktRet}_{\text{episode}}$ is the actual market return during that same recalled episode.
- $R[\text{OwnRet}_{1D}], R[\text{OwnRet}_{1M}], R[\text{OwnRet}_{1Y}], R[\text{OwnRet}_{5Y}]$ are recalled own returns over the previous day, month, year, and five years.
- $E[\text{MktRet}_{1M}]$ and $E[\text{MktRet}_{1Y}]$ are expected market returns over the next month and year.
- $E[\text{OwnRet}_{1M}]$ and $E[\text{OwnRet}_{1Y}]$ are expected own portfolio returns over the next month and year.

Interpretation. The central empirical distinction is between *actual experience* and *recalled experience*. The paper asks whether beliefs depend more on what happened or on what investors remember happening.

2.4 Summary statistics and sample facts

- The main sample has 17,324 investors: 6,214 in Neutral, 5,805 in Happy, and 5,305 in Painful.
- The merged sample has 5,154 investors with matched transaction records.
- The median respondent is about 35 years old.
- 61% have a bachelor's degree.
- 34% have wealth above 1 million RMB.
- In the introductory discussion, the paper reports that the median investor has about 36% of wealth invested in stocks, making the belief questions financially meaningful.
- In the merged sample, the average year-to-date maximum investment amount is 224,000 RMB, and average monthly turnover is 81%, indicating very active trading.

3 Conceptual framework (all equations + interpretation)

3.1 Setup: investor experiences and the cue

An investor has a database of past trading experiences. In period T , she observes current return r_T . For each past period $t = 1, \dots, T$, she has experienced return r_t . The paper simplifies by representing each experience only by its return.

Past experienced returns are modeled as draws from a normal distribution:

$$r_t \sim N(\mu, \sigma^2),$$

where μ is the objective mean return and σ^2 is return variance.

Interpretation.

- The investor has many past experiences.
- The current return r_T is a natural market cue.
- The paper asks whether the cue changes which experiences are recalled.

3.2 Free recall: similarity-based retrieval

Let $f(r)$ denote the objective density of experienced returns. If the current return r_T acts as a cue, the recalled return is drawn not from $f(r)$ directly, but from a cue-distorted distribution:

$$f^*(r \mid r_T) = f(r) \times s^*(r, r_T), \quad (1)$$

where

$$s^*(r, r_T) = \frac{s(r, r_T)}{\int_z f(z) s(z, r_T) dz}. \quad (2)$$

The similarity function is:

$$s(r, r_T) = \exp \left[-\frac{(r - r_T)^2}{2\sigma_s^2} \right]. \quad (3)$$

Interpretation.

- Experiences with returns close to the current return r_T are more similar to the cue.
- Similar experiences are more likely to be recalled.
- σ_s controls cue strength. If σ_s is very large, similarity is almost flat and the cue has little effect.

Under this specification, the expected free-recalled return is:

$$R^{free}(r \mid r_T) = (1 - \alpha)\mu + \alpha r_T, \quad \alpha = \frac{\sigma^2}{\sigma^2 + \sigma_s^2}. \quad (4)$$

Interpretation.

- The recalled return is a weighted average of the objective mean μ and the current cue r_T .
- The stronger the cue, the larger α .
- If today's market return is high, the mean recalled return is predicted to be high.

Prediction 1: cue effect on free recall. The mean of free-recalled returns is increasing in the current return r_T .

3.3 Probed recall: trying to remember a specific horizon

In ProbedRecall, the investor is asked to recall a specific experience, such as yesterday's return or last month's return. Suppose she correctly recalls the target return r_t with probability θ . With probability $1 - \theta$, retrieval fails and she falls back on cued free recall.

Then the mean probed-recalled return is:

$$R^{probed}(r_t \mid r_T) = \theta r_t + (1 - \theta) [(1 - \alpha)\mu + \alpha r_T]. \quad (5)$$

Interpretation.

- Probed recall is partly accurate: with probability θ , the target return is recovered.
- But it is still cue-contaminated: when memory search is imperfect, the current return influences what comes to mind.
- Even after controlling for actual own return r_t , current returns can bias recalled own returns.

Prediction 2: cue effect on probed recall. The recalled return for a specified past period is increasing in the current return r_T .

3.4 Simulation: from recalled experiences to beliefs

After recall, the investor forecasts future returns by simulating from retrieved experiences. In the simplest case, expected future return is the average of free-recalled returns:

$$\mathbb{E}_T[r_{T+1}] = R^{free}(r \mid r_T) = (1 - \alpha)\mu + \alpha r_T.$$

Interpretation.

- The model generates return extrapolation even when objective returns are i.i.d.

- The current return affects beliefs indirectly by cueing similar past experiences.
- A high current return produces more positive recalled samples, which produces more optimistic forecasts.

Prediction 3: return extrapolation. The investor’s expected future return is increasing in the current return r_T .

Because simulation affects the whole perceived distribution, it also affects downside-risk beliefs. Let a crash be $r_{T+1} < \bar{r}$.

Prediction 4: perceived crash risk. The perceived crash probability $\mathbb{P}(r_{T+1} < \bar{r})$ is decreasing in recalled returns and in the current return cue.

3.5 Extensions: recency, salience, and horizon dependence

The baseline model focuses on similarity-based cued recall. The paper then incorporates two additional memory forces.

Recency. Recent experiences are more likely to share context with the current state. A reduced-form recency weight is:

$$\rho(r_t) = \exp[-\sigma_r(T - t)],$$

where a larger σ_r means faster memory decay.

Salience. Extreme returns are more memorable. A salience weight is:

$$a(r_t) = \exp[\sigma_a(r_t - \mu)^2],$$

where larger σ_a makes extreme returns more likely to be recalled.

Combining similarity, recency, and salience, the retrieval probability of experience t is:

$$p_t = \frac{s(r_t, r_T)\rho(r_t)a(r_t)}{\sum_{\tau=1}^T s(r_\tau, r_T)\rho(r_\tau)a(r_\tau)}. \quad (6)$$

Interpretation.

- A recalled experience may come to mind because it is similar to the current cue.
- It may also come to mind because it is recent.
- It may also come to mind because it is extreme or vivid.
- The empirical paper looks for all three: similarity/cued recall, recency, and salience.

Horizon dependence. If investors forecast a one-year return, the forecast question itself can cue one-year memories. If they forecast a one-month return, the question can cue one-month memories. Therefore, expectations should be more related to recalls from matching horizons.

4 Empirical specifications (all equations + interpretation)

4.1 Testing cued recall in FreeRecall

The main FreeRecall specification is:

$$R[\text{MktRet}_{\text{episode}}] = \beta_0 + \beta_D \text{MktRet}_{\text{today}} + \beta_M \text{MktRet}_{1M} + \beta_Y \text{MktRet}_{1Y} + X_i' \gamma + \varepsilon_i. \quad (7)$$

Objects.

- $R[\text{MktRet}_{\text{episode}}]$ is the recalled market return during the episode that comes to mind.
- $\text{MktRet}_{\text{today}}$ is the cumulative market return from market open to the exact minute the investor starts the survey.
- MktRet_{1M} and MktRet_{1Y} are cumulative market returns over the past month and year.
- X_i contains demographic, wealth, income, account-checking, news-checking, investment-discussion, and WeChat-group controls.

Interpretation.

- $\beta_D > 0$ means today's market movement cues investors to recall more bullish episodes.
- The main FreeRecall test uses the Neutral treatment to avoid prompt-driven happiness/painfulness confounds.
- The paper excludes observations whose recalled episodes overlap with the recent return cue, reducing mechanical correlation.

4.2 Testing cued recall in ProbedRecall

The ProbedRecall specification is:

$$R[\text{OwnRet}_h] = \beta_0 + \beta_1 \text{MktRet}_{\text{today}} + X_i' \gamma + \varepsilon_i, \quad h \in \{1D, 1M\}. \quad (8)$$

In the merged sample, the authors also control for actual own returns and use actual own return today as an alternative cue:

$$R[\text{OwnRet}_h] = \beta_0 + \beta_1 \text{MktRet}_{\text{today}} + \delta \text{OwnRet}_h + X_i' \gamma + \varepsilon_i,$$

$$R[\text{OwnRet}_h] = \beta_0 + \beta_1 \text{OwnRet}_{\text{today}} + \delta \text{OwnRet}_h + X_i' \gamma + \varepsilon_i.$$

Interpretation.

- In the main sample, actual own returns are not observed, so recalled own returns are adjusted using actual market returns.
- In the merged sample, actual own returns are observed directly, so the paper can test whether current cues bias recall even after controlling for what actually happened.
- A positive β_1 supports Prediction 2.

4.3 Relating FreeRecall memories to beliefs

The FreeRecall belief specification is:

$$E[\text{Ret}] = \beta_0 + \beta_1 R[\text{MktRet}_{\text{episode}}] + X_i' \gamma + \varepsilon_i. \quad (9)$$

Interpretation.

- The dependent variable can be expected market return or expected own return.
- The forecast horizon can be one month or one year.
- $\beta_1 > 0$ means investors who recall more bullish market episodes are more optimistic about the future.

4.4 Relating ProbedRecall memories to beliefs

The ProbedRecall belief specification is:

$$E[\text{Ret}] = \beta_0 + \beta_1 \text{R}[\text{OwnRet}_{1M}] + \beta_2 \text{R}[\text{OwnRet}_{1Y}] + X'_i \gamma + \varepsilon_i. \quad (10)$$

Interpretation.

- This tests whether recalled own past performance predicts expectations.
- Horizon dependence predicts that $\text{R}[\text{OwnRet}_{1M}]$ should matter more for one-month forecasts, while $\text{R}[\text{OwnRet}_{1Y}]$ should matter more for one-year forecasts.

4.5 Crash-probability test

To test whether memories affect perceived downside risk, the paper estimates:

$$E[\text{CrashProb}] = \beta_0 + \beta_1 \text{R}[\text{OwnRet}_{1M}] + \beta_2 \text{R}[\text{OwnRet}_{1Y}] + X'_i \gamma + \varepsilon_i. \quad (11)$$

Interpretation.

- The simulation mechanism predicts $\beta_1 < 0$ and $\beta_2 < 0$.
- If investors recall higher past returns, their simulated future distribution shifts right, so perceived crash risk falls.
- This is useful against an anchoring story: mechanical anchoring on similarly worded questions would not naturally predict a negative relation with crash probabilities.

4.6 Survey-order test

To test whether the recall-belief relationship is caused by priming, the authors conduct a RandomOrder survey and estimate within each treatment:

$$E[\text{MktRet}_{1M}] = \beta_0 + \beta_1 \text{R}[\text{OwnRet}_{1M}] + X'_i \gamma + \varepsilon_i. \quad (12)$$

They also pool treatments and interact recalled returns with treatment indicators.

Interpretation.

- If the original ordering mechanically inflated the effect of recall on beliefs, the relationship should weaken when beliefs are elicited before recalls.
- The paper finds the relationship remains positive and significant under different orderings.

4.7 Trading behavior and the belief channel

The first trading regression is:

$$\text{NetBuy} = \beta_0 + \beta_1 E[\text{OwnRet}_{1M}] + \beta_2 E[\text{MktRet}_{1M}] + X_i' \gamma + \varepsilon_i. \quad (13)$$

The paper then constructs the part of own-return expectations explained by memory. First regress $E[\text{OwnRet}_{1M}]$ on $R[\text{OwnRet}_{1M}]$ and $R[\text{OwnRet}_{1Y}]$ plus controls, then call the fitted value $\hat{E}[\text{OwnRet}_{1M}]$. The trading regression is:

$$\text{NetBuy} = \beta_0 + \beta_1 \hat{E}[\text{OwnRet}_{1M}] + X_i' \gamma + \varepsilon_i. \quad (14)$$

Interpretation.

- NetBuy is buy amount minus sell amount, either on the survey day or during the following month.
- If memory affects beliefs, and beliefs affect trading, then memory-induced beliefs should predict future net buying.
- The paper finds exactly this pattern.

5 Identification and credibility (what makes the estimates credible)

5.1 Cue-to-recall link

- **Source of variation.** The survey is rolled out over six weeks. Investors take the survey on different days and at different times. Hence, they are exposed to different market-return cues.
- **Exact timing.** The survey platform records the exact minute each investor starts the survey. This allows $\text{MktRet}_{\text{today}}$ to be measured as the return up to that minute.
- **Naturalistic cue.** The survey does not explicitly ask investors about today's market return before recall. The cue is naturally present in the market environment, not experimentally inserted into the recall question.
- **Overlap controls.** In FreeRecall, the paper excludes recalled episodes that overlap with the recent cue period to avoid mechanical positive correlations.
- **Actual-return controls.** In the merged sample, actual own returns are observed, so the paper can show that current returns bias recall even after controlling for actual performance.

Credibility. These features make the cue-to-recall evidence relatively persuasive. The strongest evidence comes from ProbedRecall, where current market and own returns predict recalled past own returns despite controls for actual returns.

5.2 Recall-to-belief link

- **Main evidence.** Recalled returns are strongly related to expected returns and perceived crash probabilities.

- **Horse races.** Recalled experiences dominate actual experiences when both are included in belief regressions.
- **Horizon dependence.** Investors rely more on recalled returns from horizons that match the forecast horizon.
- **Explanatory power.** A single recalled-return variable explains about as much belief heterogeneity as a large set of demographic and social-activity variables.
- **Trading validation.** The part of beliefs explained by recall predicts future net buying, so the belief measures are not just survey noise.

Credibility. The evidence strongly supports an association between memory and beliefs. However, the paper is careful not to overclaim causality because recall and beliefs are elicited in the same survey.

5.3 Alternative explanations addressed

- **Naive anchoring on numbers.** The Happy/Painful/Neutral treatments shift recalled episode returns, but the subsequent ProbedRecall variables do not mechanically shift in the same monotonic pattern.
- **Anchoring from similar wording.** Higher recalled own returns are negatively related to perceived crash probabilities, matching the simulation model rather than a simple anchoring story.
- **Anchoring from answer formats.** Some returns are computed from RMB amounts rather than elicited directly as percentages, yet they still predict beliefs.
- **Priming.** A separate RandomOrder survey varies the order of recall and expectation blocks. The recall-belief relationship persists.
- **Cognitive defaults.** Recalled returns are positively correlated with actual returns, suggesting investors are not simply choosing default answers.
- **Motivated reasoning.** The paper checks whether prior actions predict future recall and finds little evidence that investors simply recall positive experiences to justify past trades.
- **Taste for consistency.** The paper argues that consistency is ambiguous in financial settings because both continuation and mean reversion can sound internally consistent.

5.4 Limitations

- **No exogenous manipulation of memory.** The paper does not randomly assign memories. The recall-to-belief link remains partly correlational.
- **Same-survey measurement.** Eliciting recall and beliefs in one survey may lead respondents to infer a connection between questions.
- **External information restricted.** Investors are instructed not to check accounts or search the internet. This helps elicit memory but may overstate the role of memory relative to real-world settings where investors can look up data.

- **Retail-investor setting.** The results may apply most strongly to retail investors who rely on informal impressions and less to institutional investors using formal models and research teams.
- **China-specific market.** Chinese retail investors trade frequently and may differ from investors in other markets. The mechanism is general, but magnitudes may not be.

6 Tables and figures

6.1 Figure I: chain from cue to recall, belief, and action

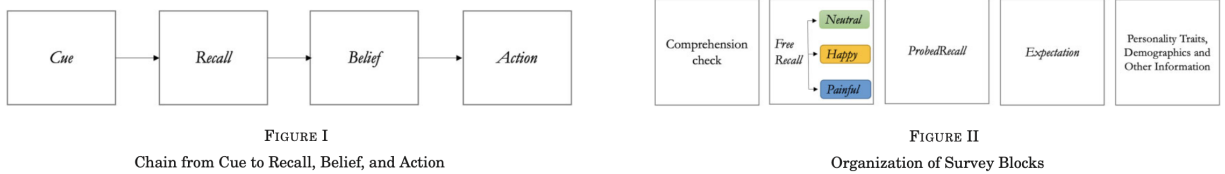
Read: This figure gives the paper’s full logic:

$$\text{cue} \longrightarrow \text{recall} \longrightarrow \text{belief} \longrightarrow \text{action}.$$

Interpretation.

- Market returns serve as cues.
- Cues affect which experiences investors recall.
- Recalled experiences shape return expectations and crash-risk beliefs.
- Beliefs affect trading decisions.

Why it matters. The paper’s empirical sections test the chain piece by piece rather than only showing one reduced-form correlation.



6.2 Figure II: organization of survey blocks

Read: The figure shows the survey order: comprehension, FreeRecall, ProbedRecall, Expectation, personality, and demographics.

Interpretation.

- FreeRecall comes before ProbedRecall so that the free recall is not contaminated by more structured questions.
- Beliefs come after recalls in the main survey, which is why anchoring and priming must be addressed later.

6.3 Figure III: demographic distributions

Read: The figure plots gender, age, income, wealth, education, and trading experience.

Key facts.

- Respondents are young: median age around 35.
- They are educated: 61% have a bachelor’s degree.
- They are relatively affluent: 34% have wealth above 1 million RMB.

Why it matters. The sample includes real investors with meaningful financial stakes, not only low-stakes survey respondents.

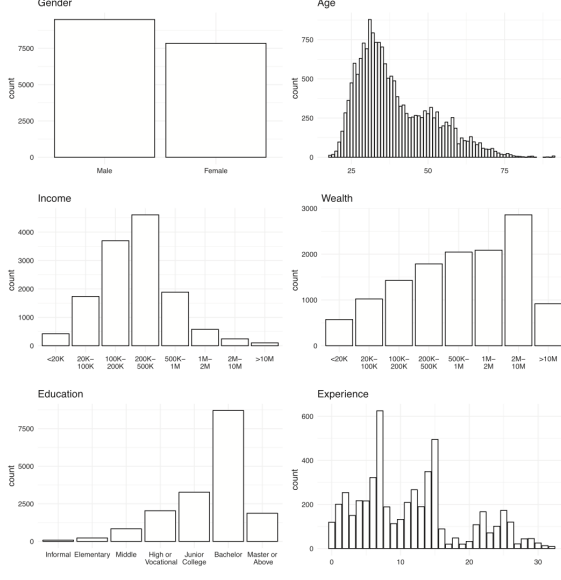


FIGURE III
Distribution of Demographic Variables

This figure reports the distribution of gender, age, annual income, wealth, education, and experience for survey respondents. The variable experience is defined as the number of years since an investor opens a trading account and is only available for the merged sample. All other variables are based on the main sample. The y-axis represents the number of observations.

TABLE I
SAMPLE CONSTRUCTION PROCESS

Filter	Sample size
Initial sample	37,921
Drop if an investor spent < 175 seconds (5th percentile) on the survey	36,164
Drop if an investor fails the two comprehension check questions	27,799
Drop if an investor's recalled episode > 10 years in <i>FreeRecall</i>	17,324
Main sample	17,324
Treatment <i>Neutral</i>	6,214
Treatment <i>Happy</i>	5,805
Treatment <i>Painful</i>	5,305
Merged sample	5,154

Notes. We survey 37,921 retail investors in China. To ensure response quality, we impose the following filters to create our main sample: (i) the time spent to complete the survey must be at least 175 seconds (5th percentile); (ii) at the beginning of the survey, respondents must correctly answer two questions related to the concept of dollar investment and dollar return; (iii) in response to the question about the episode that comes to mind in the *FreeRecall* block, the length of the recalled episode must be no longer than 10 years. After imposing these three filters, we arrive at a main sample of 17,324 investors. In the *FreeRecall* block, investors are randomly assigned to one of the three treatments—*Neutral*, *Happy*, and *Painful*—which ask respondents to recall a past market episode that first comes to mind, that is the happiest, and that is the most painful, respectively. The number of qualified responses is 6,214, 5,805, and 5,305 for the three treatments, respectively. Finally, we are able to merge the survey responses with respondents' actual trading records for a subsample of 5,154 observations. These responses constitute our merged sample.

6.4 Table I: sample construction

Read: Table I tracks how the sample goes from raw survey responses to the analysis sample.

Key numbers.

- Initial sample: 37,921.
- After time filter: 36,164.
- After comprehension checks: 27,799.
- After dropping FreeRecall episodes longer than 10 years: 17,324.
- Merged transaction sample: 5,154.

Interpretation. The strict filters sacrifice sample size to improve answer quality.

6.5 Table II: summary statistics of recalled, expected, and realized returns

Read: Table II reports recalled episode returns, probed own-return recalls, expectations, and realized own returns.

Key numbers.

- In FreeRecall, the average recalled market episode return is 4.2%, with a large standard deviation of 36.5%.
- By treatment, recalled episode returns are 5.6% in Neutral, 23.4% in Happy, and -18.1% in Painful.
- Recalled own returns are close to zero at short horizons but more positive at longer horizons: mean recalled own return is -0.3% for one day, -0.2% for one month, 1.8% for one year, and 4.3% for five years.
- Expected own returns are much higher than expected market returns: mean $E[\text{OwnRet}_{1Y}]$ is 16.0%, while mean $E[\text{MktRet}_{1Y}]$ is 5.2%.

Interpretation. Investors are heterogeneous in memory and beliefs. They also appear overconfident about their own future returns relative to market returns.

TABLE II
SUMMARY STATISTICS OF RECALLED AND REALIZED RETURNS

	<i>N</i>	Mean	Std. dev.	P5	P25	Median	P75	P95
Panel A: Main sample, the <i>FreeRecall</i> block								
$\mathbb{R}[\text{MktRet}_{\text{episode}}]$	13,791	4.2%	36.5%	-50.5%	-17.5%	0.0%	12.5%	100.0%
<i>Neutral</i>	5,087	5.6%	38.8%	-50.5%	-19.5%	0.0%	15.5%	100.0%
<i>Happy</i>	4,511	23.4%	33.6%	-8.5%	2.5%	9.5%	30.5%	100.0%
<i>Painful</i>	4,193	-18.1%	21.3%	-50.5%	-32.5%	-15.5%	-3.5%	8.5%
Panel B: Main sample, the <i>ProbedRecall</i> block								
$\mathbb{R}[\text{OwnRet}_{1D}]$	10,432	-0.3%	5.5%	-13.5%	-2.5%	-0.5%	2.5%	10.5%
$\mathbb{R}[\text{OwnRet}_{1M}]$	9,957	-0.2%	6.5%	-13.5%	-4.5%	0.5%	4.5%	10.5%
$\mathbb{R}[\text{OwnRet}_{1Y}]$	10,440	1.8%	13.2%	-22.5%	-6.5%	1.5%	8.5%	32.5%
$\mathbb{R}[\text{OwnRet}_{5Y}]$	9,325	4.3%	24.3%	-39.5%	-9.5%	2.5%	10.5%	70.5%
Panel C: Main sample, the <i>Expectation</i> block								
$E[\text{MktRet}_{1M}]$	12,786	2.0%	3.8%	-5.5%	0.0%	1.5%	4.5%	9.5%
$E[\text{MktRet}_{1Y}]$	12,356	5.2%	6.7%	-6.5%	0.0%	4.5%	9.5%	20.5%
$E[\text{OwnRet}_{1M}]$	9,375	6.1%	7.7%	-5.5%	2.5%	4.5%	9.5%	28.5%
$E[\text{OwnRet}_{1Y}]$	9,602	16.0%	23.4%	-5.5%	4.5%	9.5%	19.5%	99.4%
Panel D: Merged sample								
OwnRet_{1D}	2,759	0.3%	2.5%	-3.0%	-0.9%	0.2%	1.4%	4.0%
OwnRet_{1M}	2,936	3.2%	10.1%	-10.5%	-2.0%	2.6%	7.3%	18.6%
OwnRet_{1Y}	3,309	8.0%	28.0%	-24.1%	-6.8%	3.5%	16.9%	52.1%

Notes. This table reports summary statistics for recalled and actual experiences in the *FreeRecall* and *ProbedRecall* blocks. In the *FreeRecall* block, investors are randomly assigned to one of the three treatments, *Neutral*, *Happy*, or *Painful*, which ask respondents to recall a past market episode that first comes to mind, that is the happiest, and that is the most painful, respectively. Respondents are then asked to recall the market return, labeled $\mathbb{R}[\text{MktRet}_{\text{episode}}]$ during the recalled episode. The *ProbedRecall* block asks investors to recall returns of their own portfolios over the past one day, one month, one year, and five years, labeled as $\mathbb{R}[\text{OwnRet}_{1D/1M/1Y/5Y}]$. The *Expectation* block asks investors to predict future returns of the market and their own portfolios in the next month and next year, labeled as $E[\text{MktRet}_{1M/1Y}]$ and $E[\text{OwnRet}_{1M/1Y}]$. In Panels A–C, we report the summary statistics of these variables in the main sample. For a subsample of respondents for whom we can observe their transactions (the merged sample), we also calculate their *actual* portfolio returns over the same period of time, denoted by $\text{OwnRet}_{1D/1M/1Y/5Y}$, and report the summary statistics in Panel D.

TABLE III
CORRELATIONS BETWEEN RECALLED AND ACTUAL RETURNS

	$\mathbb{R}[\text{MktRet}_{\text{episode}}]$	$\mathbb{R}[\text{OwnRet}_{1D}]$	$\mathbb{R}[\text{OwnRet}_{1M}]$	$\mathbb{R}[\text{OwnRet}_{1Y}]$
$\text{MktRet}_{\text{episode}}$	0.57*** (0.01)			
OwnRet_{1D}		0.10*** (0.02)		
OwnRet_{1M}			0.25*** (0.02)	
OwnRet_{1Y}				0.32*** (0.02)

Notes. This table reports the correlation coefficients between recalled and corresponding actual returns in the *FreeRecall* for the main sample and in the *ProbedRecall* for the merged sample. For example, the top left cell shows the correlation coefficient between the recalled and actual market episode return ($\mathbb{R}[\text{MktRet}_{\text{episode}}]$ and $\text{MktRet}_{\text{episode}}$). Standard errors are in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

6.6 Table III: correlations between recalled and actual returns

Read: Table III validates the recall measures by comparing recalled returns to actual returns.

Key numbers.

- Correlation between recalled and actual market episode return in FreeRecall: 0.57.

- Correlation between recalled and actual own return in ProbedRecall: 0.10 for one day, 0.25 for one month, and 0.32 for one year.

Interpretation. Investors are not inventing random memories. But the correlations are far below one, leaving room for biased and selective recall.

6.7 Figure IV: distribution of recalled market episodes

Read: The figure overlays recalled episode start/end months with the Shanghai Composite Index.

Main patterns.

- **Recency:** recent episodes are recalled disproportionately often.
- **Salience:** investors also recall dramatic market episodes, especially the 2007–2008 and 2014–2015 bubble/crash periods.

Interpretation. Memory is not simply a time-decaying record of experience. It oversamples both recent experiences and vivid extreme episodes.

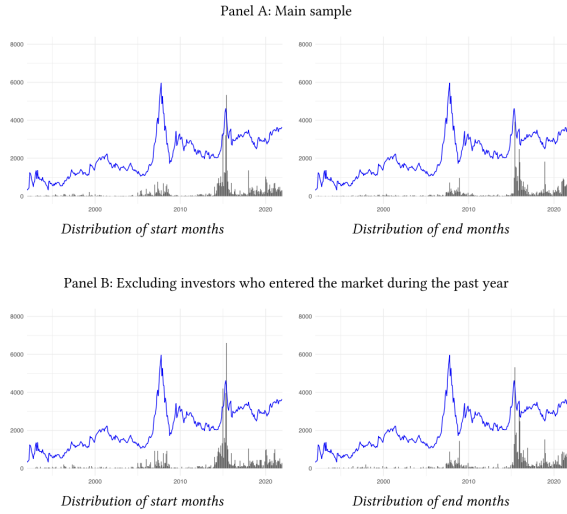


FIGURE IV

Distribution of Recalled Market Episodes in *FreeRecall*

In the *Neutral* treatment of the *FreeRecall* block, investors are asked to recall a past market episode that first comes to mind. We plot the distribution of start and end months of the reported episodes. The solid blue line represents the Shanghai Composite Index. The solid black bars represent the frequency of responses. The legends on the y-axes represent the level of the Shanghai Composite Index. Panel A is based on the main sample. Panel B is based on a subsample that excludes investors who started trading stocks during the last year of our sample (2021).

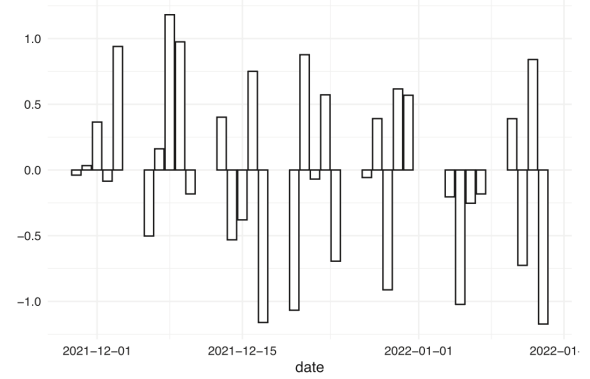


FIGURE V

Distribution of Daily Returns During the Survey Period

The figure plots the daily stock market return (in %) for the Shanghai Composite Index during our survey period, November 29, 2021, to January 9, 2022.

6.8 Figure V: daily returns during the survey period

Read: Figure V plots daily Shanghai Composite returns during the survey period.

Key numbers.

- Maximum daily return: 1.18%.
- Minimum daily return: -1.16%.

- Standard deviation: 0.66%.

Interpretation. The survey period has enough return variation to test cue effects, but not extreme market turbulence. This helps explain why FreeRecall cue effects are weaker in the full sample.

6.9 Table IV: tests of cued recall in FreeRecall

Read: Table IV regresses recalled market episode returns on current and recent market returns.

Key results.

- In the full Neutral-treatment sample, coefficients on $\text{MktRet}_{\text{today}}$, MktRet_{1M} , and MktRet_{1Y} are close to zero and insignificant.
- In the recent-recall sample, the cue effect becomes positive. A 1 percentage point increase in $\text{MktRet}_{\text{today}}$ is associated with a 2.49 to 4.09 percentage point increase in recalled episode return, depending on specification.
- MktRet_{1M} is also positive in the recent-recall sample: 0.94 in one specification and 1.63 in the joint specification.

Interpretation. Current market returns can cue FreeRecall, but the effect is easier to detect for episodes closer to the current context. Deep, salient memories may be less fluid.

TABLE IV
TESTS OF CUED RECALL IN THE *FreeRecall* BLOCK

	Dependent variable: $\mathbb{R}[\text{MktRet}_{\text{episode}}]$							
	Main sample				Sample of recent recalls			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$\text{MktRet}_{\text{today}}$	0.36 (1.35)			0.74 (1.76)	2.49** (1.24)			4.09** (1.60)
MktRet_{1M}		-0.51 (0.57)		-0.08 (0.65)		0.94* (0.48)		1.63*** (0.60)
MktRet_{1Y}			1.25 (1.21)	1.32 (1.27)			-1.43 (1.19)	0.25 (0.86)
Observations	3,503	3,503	3,503	3,503	940	940	940	940
Adjusted R^2	0.01	0.01	0.01	0.01	0.02	0.02	0.01	0.02

Notes. We test the effects of cues on *FreeRecall* by regressing recalled market return for the episode, $\mathbb{R}[\text{MktRet}_{\text{episode}}]$, on cues. We consider three types of cues: $\text{MktRet}_{\text{today}}$ is the market return on the day when the survey was completed and is calculated as the cumulative return from the market open to the minute when the investor starts to take the survey; MktRet_{1M} (MktRet_{1Y}) is the market return over the past month (year) before the survey was taken. All results are produced using the *Neutral* treatment. We also exclude observations in which the recalled episode ends in or after December 2020, so that the recalled episode does not overlap with the cues. Columns (1)–(4) are based on the entire *Neutral* treatment; columns (5)–(8) are based on a subset of the *Neutral* treatment in which the recalled episode ending date is within the past one to five years. We control for age, gender, education, wealth, income, frequency of checking accounts, frequency of checking news, frequency of discussing investments, and number of WeChat groups. Standard errors, in parentheses, are clustered by date. * $p < .1$; ** $p < .05$; *** $p < .01$.

TABLE V
TESTS OF CUED RECALL IN THE *ProbedRecall* BLOCK

	Main sample		Merged sample			
	$\mathbb{R}[\text{OwnRet}_{1D}]$	$\mathbb{R}[\text{OwnRet}_{1M}]$	$\mathbb{R}[\text{OwnRet}_{1D}]$	$\mathbb{R}[\text{OwnRet}_{1M}]$		
	(1)	(2)	(3)	(4)	(5)	(6)
$\text{MktRet}_{\text{today}}$	0.52** (0.22)	1.13*** (0.26)	1.04*** (0.33)		0.90* (0.52)	
$\text{OwnRet}_{\text{today}}$				0.28** (0.11)		0.39*** (0.10)
OwnRet_{1D}			0.35*** (0.11)	0.31*** (0.11)		
OwnRet_{1M}					0.20*** (0.02)	0.20*** (0.02)
Observations	7,362	7,362	1,348	1,348	1,348	1,348
Adjusted R^2	0.03	0.04	0.06	0.05	0.09	0.10

Notes. We test the effects of cues on *ProbedRecall* by regressing recalled own return yesterday or the last month, $\mathbb{R}[\text{OwnRet}_{1D/1M}]$, on return cues. We consider two types of cues. In columns (1) and (2), which use the main sample, the cue is $\text{MktRet}_{\text{today}}$, the market return on the day when the survey was completed, calculated as the cumulative return from the market opening to the minute when the investor begins the survey. In columns (3)–(6), which uses the merged sample, we consider a second cue, $\text{OwnRet}_{\text{today}}$, which represents an investor's actual portfolio return on the survey day. In the merged sample, we control for actual portfolio return yesterday or the past month, $\text{OwnRet}_{1D/1M}$. For all columns, we control for age, gender, education, wealth, income, frequency of checking accounts, frequency of checking news, frequency of discussing investments, and number of WeChat groups. Standard errors, in parentheses, are clustered by date, and we include treatment fixed effects. * $p < .1$; ** $p < .05$; *** $p < .01$.

6.10 Table V: tests of cued recall in ProbedRecall

Read: Table V regresses recalled own returns on current market or own-return cues.

Key results.

- Main sample: a 1 percentage point increase in $\text{MktRet}_{\text{today}}$ increases recalled previous-day own return by 0.52 percentage points and recalled past-month own return by 1.13 percentage points.
- Merged sample with actual-return controls: $\text{MktRet}_{\text{today}}$ increases recalled previous-day own return by 1.04 percentage points and recalled past-month own return by 0.90 percentage points.

- Using own return today as the cue, $\text{OwnRet}_{\text{today}}$ has coefficients 0.28 for recalled one-day own return and 0.39 for recalled one-month own return.

Interpretation. ProbedRecall is strongly cue-dependent. Investors’ memories of their own recent performance shift with today’s market and portfolio performance.

6.11 Table VI: relationship between recalls and beliefs

Read: Table VI regresses expected returns on recalled returns.

Panel A: FreeRecall.

- $R[\text{MktRet}_{\text{episode}}]$ is positively related to all four expectation measures.
- Coefficients are 0.005 for $E[\text{MktRet}_{1M}]$, 0.01 for $E[\text{MktRet}_{1Y}]$, 0.01 for $E[\text{OwnRet}_{1M}]$, and 0.05 for $E[\text{OwnRet}_{1Y}]$.

Panel B: ProbedRecall.

- $R[\text{OwnRet}_{1M}]$ has coefficients 0.09, 0.08, 0.20, and 0.16 across the four belief measures.
- $R[\text{OwnRet}_{1Y}]$ has coefficients 0.02, 0.05, 0.10, and 0.38.

Interpretation.

- Investors who recall better past performance are more optimistic about future returns.
- The horizon-dependence pattern appears: one-month recalls matter more for one-month forecasts, and one-year recalls matter more for one-year forecasts.
- Economically, moving recalled past one-month own returns from the 25th to 75th percentile raises return expectations by about 0.7 to 1.8 percentage points.

6.12 Table VII: horse race between actual and recalled experience

Read: Table VII includes actual and recalled experiences in the same regressions.

Panel A: FreeRecall.

- Actual episode market return is insignificant or negative.
- Recalled episode market return remains positive and significant.

Panel B: ProbedRecall, main sample.

- Actual past market returns initially look like extrapolation inputs.
- Once recalled own returns are added, actual market returns lose significance.
- Recalled own returns remain positive and significant.

Panel C: ProbedRecall, merged sample.

- Actual own returns lose explanatory power when recalled own returns are included.
- Recalled own returns remain strongly related to own-return expectations.

TABLE VI
RELATIONSHIP BETWEEN RECALLS AND BELIEFS

	E[MktRet _{1M}] (1)	E[MktRet _{1Y}] (2)	E[OwnRet _{1M}] (3)	E[OwnRet _{1Y}] (4)
Panel A: <i>FreeRecall</i>				
R[MktRet _{episode}]	0.005** (0.002)	0.01*** (0.003)	0.01*** (0.004)	0.05*** (0.01)
Observations	5,626	5,626	5,626	5,626
Adjusted R^2	0.01	0.04	0.04	0.07
Panel B: <i>ProbedRecall</i>				
R[OwnRet _{1M}]	0.09*** (0.01)	0.08*** (0.02)	0.20*** (0.03)	0.16* (0.09)
R[OwnRet _{1Y}]	0.02*** (0.01)	0.05*** (0.01)	0.10*** (0.01)	0.38*** (0.03)
Observations	4,190	4,190	4,190	4,190
Adjusted R^2	0.04	0.05	0.13	0.12

Notes. We explore the relationship between recalls, elicited in the *FreeRecall* and *ProbedRecall* blocks, and beliefs, elicited in the *Expectation* block. The dependent variables are four measures of beliefs: the expected return (E) of the market (MktRet) or the investor's own portfolio (OwnRet) over the next month (1M) or year (1Y). In Panel A, the key independent variable is the recalled market return for the episode, R[MktRet_{episode}], as elicited in *FreeRecall*. To ensure that coefficients are comparable across columns, we exclude investors with any missing values. In Panel B, the independent variables are the recalled own return for either the past month or year, R[OwnRet_{1M/1Y}], as elicited in the *ProbedRecall* block. Both panels use the main sample. We control for age, gender, education, wealth, income, frequency of account checks, frequency of news checks, frequency of investment discussions, and the number of WeChat groups. Standard errors, in parentheses, are clustered by date, and we include treatment fixed effects. * $p < .1$; ** $p < .05$; *** $p < .01$.

TABLE VII
HORSE RACE BETWEEN ACTUAL AND RECALLED EXPERIENCE IN EXPLANATORY POWER FOR BELIEFS

	(1)	(2)	(3)	(4)
Panel A: <i>FreeRecall</i> , main sample				
	E[MktRet _{1M}]	E[MktRet _{1Y}]	E[OwnRet _{1M}]	E[OwnRet _{1Y}]
MktRet _{episode}	-0.001 (0.001)	-0.002 (0.004)	-0.01*** (0.003)	-0.02*** (0.01)
R[MktRet _{episode}]	0.005*** (0.002)	0.02*** (0.004)	0.01*** (0.004)	0.06*** (0.02)
Observations	4,977	4,977	4,977	4,977
Adjusted R^2	0.01	0.04	0.04	0.07
Panel B: <i>ProbedRecall</i> , main sample				
	E[MktRet _{1M}]		E[OwnRet _{1Y}]	
MktRet _{1M}	0.12* (0.06)	0.09 (0.06)	0.14** (0.07)	0.10 (0.07)
MktRet _{1Y}	-0.05 (0.17)	-0.003 (0.17)	-0.19 (0.15)	-0.16 (0.15)
R[OwnRet _{1M}]		0.07*** (0.01)		0.07*** (0.02)
R[OwnRet _{1Y}]		0.01*** (0.004)		0.05*** (0.01)
Observations	6,436	6,436	6,287	6,287
Adjusted R^2	0.02	0.04	0.04	0.06
Panel C: <i>ProbedRecall</i> , merged sample				
	E[OwnRet _{1M}]		E[OwnRet _{1Y}]	
	(1)	(2)	(3)	(4)
OwnRet _{1M}	0.05* (0.03)	0.01 (0.03)	0.11 (0.07)	0.05 (0.07)
OwnRet _{1Y}	0.001 (0.02)	-0.03** (0.02)	0.05 (0.04)	-0.04 (0.04)
R[OwnRet _{1M}]		0.18*** (0.06)		0.26 (0.18)
R[OwnRet _{1Y}]		0.11*** (0.03)		0.32*** (0.09)
Observations	1,088	1,088	1,159	1,159
Adjusted R^2	0.03	0.10	0.03	0.08

Notes. We regress measures of subjective beliefs on recalled and actual experiences simultaneously. The dependent variables are four measures of beliefs: the return expectation (E) for the market return (MktRet) or the investor's own portfolio return (OwnRet) over the next month (1M) or year (1Y). In Panel A, the independent variables are the recalled market return for the episode, R[MktRet_{episode}], and the corresponding actual market return, MktRet_{episode}. In Panel B, the independent variables are the recalled own returns for the past month or year, R[OwnRet_{1M/1Y}], and the actual market returns over the same periods, MktRet_{1M/1Y}. In Panel C, the independent variables are the recalled own returns for the past month or year, R[OwnRet_{1M/1Y}], and the actual own returns over the same periods, OwnRet_{1M/1Y}. Panels A and B are based on the main sample, and Panel C is based on the merged sample. We control for age, gender, education, wealth, income, frequency of checking accounts, frequency of checking news, frequency of discussing investments, and number of WeChat groups. Standard errors, in parentheses, are clustered by date, and treatment fixed effects are included. * $p < .1$; ** $p < .05$; *** $p < .01$.

Interpretation. Beliefs are closer to subjective memories than to objective histories. This is one of the paper’s most important findings.

6.13 Table VIII: explanatory power for cross-sectional variation in beliefs

Read: Table VIII reports adjusted R^2 values from belief regressions using demographics, expanded demographics, or recalled own returns alone.

Key numbers.

- For $E[\text{MktRet}_{1M}]$, recalled own return alone gives adjusted $R^2 = 0.022$, compared with 0.009 for demographics and 0.020 for expanded demographics.
- For $E[\text{OwnRet}_{1M}]$, recalled own return alone gives adjusted $R^2 = 0.080$, compared with 0.030 for demographics and 0.047 for expanded demographics.
- For $E[\text{OwnRet}_{1Y}]$, recalled own return alone gives adjusted $R^2 = 0.073$, compared with 0.044 for demographics and 0.068 for expanded demographics.

Interpretation. A single memory variable explains belief heterogeneity about as well as, and often better than, rich demographic controls.

	$E[\text{MktRet}_{1M}]$	$E[\text{MktRet}_{1Y}]$	$E[\text{OwnRet}_{1M}]$	$E[\text{OwnRet}_{1Y}]$
	(1)	(2)	(3)	(4)
Demographics fixed effects only	0.009	0.025	0.030	0.044
Expanded demographics fixed effects	0.020	0.043	0.047	0.068
Recalled own return only	0.022	0.025	0.080	0.073

Notes. We regress measures of subjective beliefs on either demographic variables or recalled own returns. Each cell reports the adjusted R^2 from regressing expected returns on either recalled own returns alone or demographic fixed effects alone. The dependent variables are four measures of beliefs: the return expectation (E) for the market (MktRet) or the investor’s own portfolio (OwnRet) over the next month (1M) or year (1Y). In the first row, we report the adjusted R^2 from regressing a measure of belief on demographic fixed effects, including gender, age, income, wealth, and education. In the second row, we repeat this exercise, adding four additional measures of investors’ social activities: frequency of checking stock accounts, frequency of checking news, frequency of discussing investments, and the number of WeChat groups. In the third row, we report the adjusted R^2 from regressing the measure of belief on the recalled own returns over the past period of the same length. For example, the independent variable is the recalled one-month own return ($R[\text{OwnRet}_{1M}]$) if the dependent variable is the expectation of future one-month market or own returns ($E[\text{MktRet}_{1M}]$ or $E[\text{OwnRet}_{1M}]$).

	(1)	(2)	(3)	(4)	(5)
Panel A: Testing naive anchoring on numbers					
	$R[\text{MktRet}_{\text{episode}}]$	$R[\text{OwnRet}_{1D}]$	$R[\text{OwnRet}_{1M}]$	$R[\text{OwnRet}_{1Y}]$	$R[\text{OwnRet}_{5Y}]$
<i>Neutral</i>	5.6%	−0.3%	−0.1%	2.1%	4.9%
<i>Happy</i>	23.4%	−0.2%	−0.4%	1.8%	5.1%
<i>Painful</i>	−18.1%	−0.5%	−0.1%	1.6%	2.8%
Panel B: Testing anchoring due to similarity in question phrasing					
	$R[\text{CrashProb}_{1M}]$		$R[\text{CrashProb}_{1Y}]$		
$R[\text{OwnRet}_{1M}]$	−0.11*** (0.02)	−0.08*** (0.02)		−0.05*** (0.01)	
$R[\text{OwnRet}_{1Y}]$		−0.03*** (0.01)	−0.03*** (0.01)	−0.02*** (0.01)	
Observations	7,853	7,853	7,832	7,832	
Adjusted R^2	0.02	0.02	0.03	0.03	
Panel C: Testing anchoring due to similarity in answer options					
	$E[\text{MktRet}_{1M}]$	$E[\text{MktRet}_{1Y}]$	$E[\text{OwnRet}_{1M}]$	$E[\text{OwnRet}_{1Y}]$	
$R[\text{OwnRet}_{\text{episode}}]$	0.004*** (0.001)	0.01*** (0.002)	0.01*** (0.002)	0.04*** (0.01)	
Observations	5,518	5,518	5,518	5,518	
Adjusted R^2	0.01	0.04	0.04	0.06	

Notes. Each panel tests a different type of anchoring: naive anchoring on numbers, anchoring due to similarity in question phrasing, and anchoring due to similarity in answer options. Panel A tests for naive anchoring on numbers by reporting the mean values of recalled market returns for the episode, $R[\text{MktRet}_{\text{episode}}]$, and recalled own returns from the past, $R[\text{OwnRet}_{1D/1M/1Y/5Y}]$, across three treatments: *Neutral*, *Happy*, and *Painful*. In Panel B, we test for anchoring due to similarity in question phrasing by regressing expected crash probability on recalled own returns from the past month or year. During the sample period, the Shanghai Composite Index mostly hovered between 3,500 and 3,600. We consider two potential crash events: the index dropping below 3,000 within a month and the index dropping below 2,500 within a year. Investors are asked to report a percentage between 0% and 100% as their subjective probability of a crash, $E[\text{CrashProb}_{1M/1Y}]$. In Panel C, we regress measures of subjective beliefs on recalled episode own returns. The dependent variables are four measures of beliefs: the return expectation (E) for the market return (MktRet) or the investor’s own portfolio return (OwnRet) over the next month (1M) or year (1Y). The main independent variable is the recalled own return for the episode, $R[\text{OwnRet}_{\text{episode}}]$. This is elicited by first asking about the total RMB investment and then the total RMB return; the ratio of the RMB return to the investment is the return rate. In Panels B and C, we control for age, gender, education, wealth, income, frequency of checking accounts, frequency of checking news, frequency of discussing investments, and number of WeChat groups. Standard errors, in parentheses, are clustered by date, and we include treatment fixed effects. * $p < .1$; ** $p < .05$; *** $p < .01$.

6.14 Table IX: tests for anchoring in simulation

Read: Table IX checks whether the recall-belief relationship is driven by mechanical anchoring.

Panel A: naive anchoring on numbers.

- FreeRecall treatment means are very different: Neutral 5.6%, Happy 23.4%, Painful -18.1%.
- But ProbedRecall own returns do not move monotonically in the same way across treatments.

Panel B: similarity in question phrasing.

- Recalled own returns are negatively related to expected crash probabilities.
- For example, $R[\text{OwnRet}_{1M}]$ has coefficient -0.11 on one-month crash probability and -0.05 on one-year crash probability in the joint specification.

Panel C: similarity in answer options.

- Recalled episode own return, computed from RMB amounts, still predicts return expectations.

Interpretation. The results are more consistent with simulation from recalled experiences than with simple anchoring across survey answers.

6.15 Table X: testing order effects in simulation

Read: Table X uses an additional RandomOrder survey with three orders: Baseline, Reverse, and Switch.

Key results.

- In Panel A, the coefficient of $R[\text{OwnRet}_{1M}]$ on $E[\text{MktRet}_{1M}]$ is positive in all three orders: 0.13 in Baseline, 0.16 in Reverse, and 0.10 in Switch.
- In Panel B, interaction terms with Reverse and Switch are generally insignificant.
- The sample has 1,202 investors after filters.

Interpretation. Changing survey order does not remove the recall-belief relationship. Priming from recall-first survey order is unlikely to be the sole explanation.

6.16 Table XI: beliefs, recalls, and trading decisions

Read: Table XI tests whether survey beliefs and memory-induced beliefs predict future net buying.

Key results.

- A 1 percentage point increase in $E[\text{OwnRet}_{1M}]$ is associated with 26 RMB more net buying on the survey day and 296 RMB more net buying during the following month.
- Expected market return is not statistically significant once expected own return is included.
- A 1 percentage point increase in memory-induced $\hat{E}[\text{OwnRet}_{1M}]$ is associated with 71 RMB more net buying on the survey day and 1,141 RMB more net buying during the following month.
- These effects remain after controlling for actual past own return.

Interpretation. Recalled experiences are not just survey artifacts. They are linked to beliefs that predict actual trading behavior.

TABLE X
TESTING ORDER EFFECTS IN SIMULATION

	(1)	(2)	(3)
Panel A: Regressing beliefs on recalls by treatment			
	E[MktRet _{1M}]		
	<i>Baseline</i>	<i>Reverse</i>	<i>Switch</i>
R[OwnRet _{1M}]	0.13*** (0.03)	0.16*** (0.03)	0.10** (0.04)
Observations	395	429	378
Adjusted R ²	0.04	0.05	0.01
Panel B: Statistical tests of order effects			
	E[MktRet _{1M}]	E[OwnRet _{1M}]	E[CrashProb _{1M}]
R[OwnRet _{1M}]	0.15*** (0.03)	0.24*** (0.06)	-0.45*** (0.17)
<i>Reverse</i>	-0.001 (0.33)	-0.42** (0.21)	-0.71 (1.03)
<i>Switch</i>	-0.22 (0.31)	-0.60** (0.24)	1.15 (1.00)
R[OwnRet _{1M}] × <i>Reverse</i>	0.02 (0.02)	0.22*** (0.07)	-0.11 (0.17)
R[OwnRet _{1M}] × <i>Switch</i>	-0.06 (0.06)	0.09 (0.13)	0.35 (0.40)
Observations	1,202	1,202	1,202
Adjusted R ²	0.04	0.12	0.01

Notes. To examine the effect of question ordering on the statistical relationship between recalls and beliefs, we conducted a new survey that varies the order of survey blocks. The first treatment, *Baseline*, follows the same order as our baseline survey: *FreeRecall*, *ProbedRecall*, and *Expectation*. The second treatment, *Reverse*, reverses the order: *Expectation*, *FreeRecall*, and *ProbedRecall*. The third treatment, *Switch*, swaps the order between probed recall and beliefs: *FreeRecall*, *Expectation*, and *ProbedRecall*. Each respondent was randomly assigned to one of the three treatments. In Panel A, we regress the expected market return for the next month (E[MktRet_{1M}]) on the recalled one-month own return (R[OwnRet_{1M}]) separately for the three treatments. In Panel B, we pool all treatments together and regress expected future returns or crash probability on recalled one-month own returns, along with two dummies indicating the *Reverse* and *Switch* treatments, and interaction terms between the recalled one-month own return and the two dummies. We control for age, gender, education, wealth, income, frequency of checking accounts, frequency of checking news, frequency of discussing investments, and the number of WeChat groups. Standard errors, in parentheses, are clustered by date. * $p < .1$; ** $p < .05$; *** $p < .01$.

TABLE XI
BELIEFS, RECALLS, AND TRADING DECISIONS

	(1)	(2)	(3)
Panel A: Dependent variable NetBuy _{1D}			
E[OwnRet _{1M}]	26.15* (14.94)		
E[MktRet _{1M}]	-18.72 (22.95)		
$\tilde{E}$ [OwnRet _{1M}]		71.16*** (24.49)	79.67** (37.52)
OwnRet _{1M}			13.58 (9.58)
Observations	1,869	1,783	1,088
Adjusted R ²	0.01	0.01	0.00
Panel B: Dependent variable NetBuy _{1M}			
E[OwnRet _{1M}]	295.52*** (146.90)		
E[MktRet _{1M}]	219.51 (368.17)		
$\tilde{E}$ [OwnRet _{1M}]		1,141.48*** (434.28)	1,347.80** (590.62)
OwnRet _{1M}			243.93 (158.27)
Observations	1,869	1,783	1,088
Adjusted R ²	-0.01	0.00	-0.01

Notes. We regress measures of future trading behavior on return expectations and actual experiences. The dependent variables are NetBuy_{1D/1M}, the net buy amount (buy minus sell) on the day of the survey or in the month after the survey. The independent variables are E[OwnRet_{1M}] and E[MktRet_{1M}], return expectations of own portfolios or the market over the next month; $\tilde{E}$ [OwnRet_{1M}], return expectation of own portfolio return in the next month that can be explained by recalled own returns; and OwnRet_{1M}, actual own return over the past month. The test is based on the merged sample. We control for age, gender, education, wealth, income, frequency of checking accounts, frequency of checking news, frequency of discussing investments, and number of WeChat groups. Standard errors, in parentheses, are clustered by date. * $p < .1$; ** $p < .05$; *** $p < .01$.

7 Short summary

- The paper studies how retail investors remember past stock-market experiences and how these memories shape beliefs.
- The model has two steps: similarity-based recall and simulation-based belief formation.
- The survey elicits FreeRecall, ProbedRecall, return expectations, crash probabilities, demographics, and personality/social variables from over 17,000 Chinese retail investors.
- For more than 5,000 investors, survey data are merged to actual transaction records.
- Recalled returns are correlated with actual returns, validating the recall measures.
- Memory is selective: investors recall recent episodes and salient bubble/crash episodes.
- Memory is cued: current market and own returns affect recalled past own returns.
- Recalled experiences strongly explain return expectations and crash-risk beliefs.
- In horse races, recalled experiences matter more than actual experiences.
- Memory-induced beliefs predict net buying after the survey.
- The main limitation is that the recall-to-belief relationship is not fully causal because recall and belief are measured in survey data.

8 Conclusion

8.1 Core conclusion

The paper shows that investor memory is a central input into belief formation. Retail investors do not simply remember the past accurately and then forecast rationally. Instead, they recall selectively, influenced by similarity, recency, and salience. These recalled experiences shape return expectations, perceived crash risk, and trading decisions.

8.2 Economic intuition

A high current return makes positive past experiences easier to retrieve. Once retrieved, those experiences become the mental sample from which investors simulate the future. The investor then feels as though future returns are likely to be higher and crash risk lower. This process creates extrapolative beliefs even without objective return predictability.

8.3 Takeaway

The paper provides field evidence for a memory-based theory of financial beliefs. The most important takeaway is not merely that investors extrapolate. It is that extrapolation may come from the way memory works: investors forecast the future using a selective and cue-dependent sample of the past.